# Cyber Security Assessment Report – Task 1

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**Track:** Cyber Security
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## 1. Executive Summary

This report presents a read-only security assessment of the publicly accessible website: https://example.com.

The purpose is to identify potential security weaknesses, classify the risk levels, and provide business-friendly remediation recommendations that can be implemented practically.

The analysis used passive techniques only; no attacks or unauthorized access were performed.

## 2. Scope of Assessment

**Website Tested:** https://example.com

**Assessment Scope:**

- Publicly accessible pages only
- Passive, read-only analysis
- No exploitation, login, or restricted area access

**Tools Used:**

- Browser DevTools (Inspect Element & Network)
- SecurityHeaders.com
- BuiltWith.com

## 3. Methodology

1. **Passive Observation**
   - Browsed the website and inspected HTML, network requests, and headers using DevTools.
2. **Header Analysis**
   - Checked for security headers such as Content-Security-Policy, X-Frame-Options, Strict-Transport-Security, and X-Content-Type-Options.
3. **Technology Stack Identification**
   - Used BuiltWith.com to identify frameworks, libraries, and components.

4. **Risk Classification**
    - Findings were classified as **Low, Medium, or High** risk based on potential impact and likelihood.

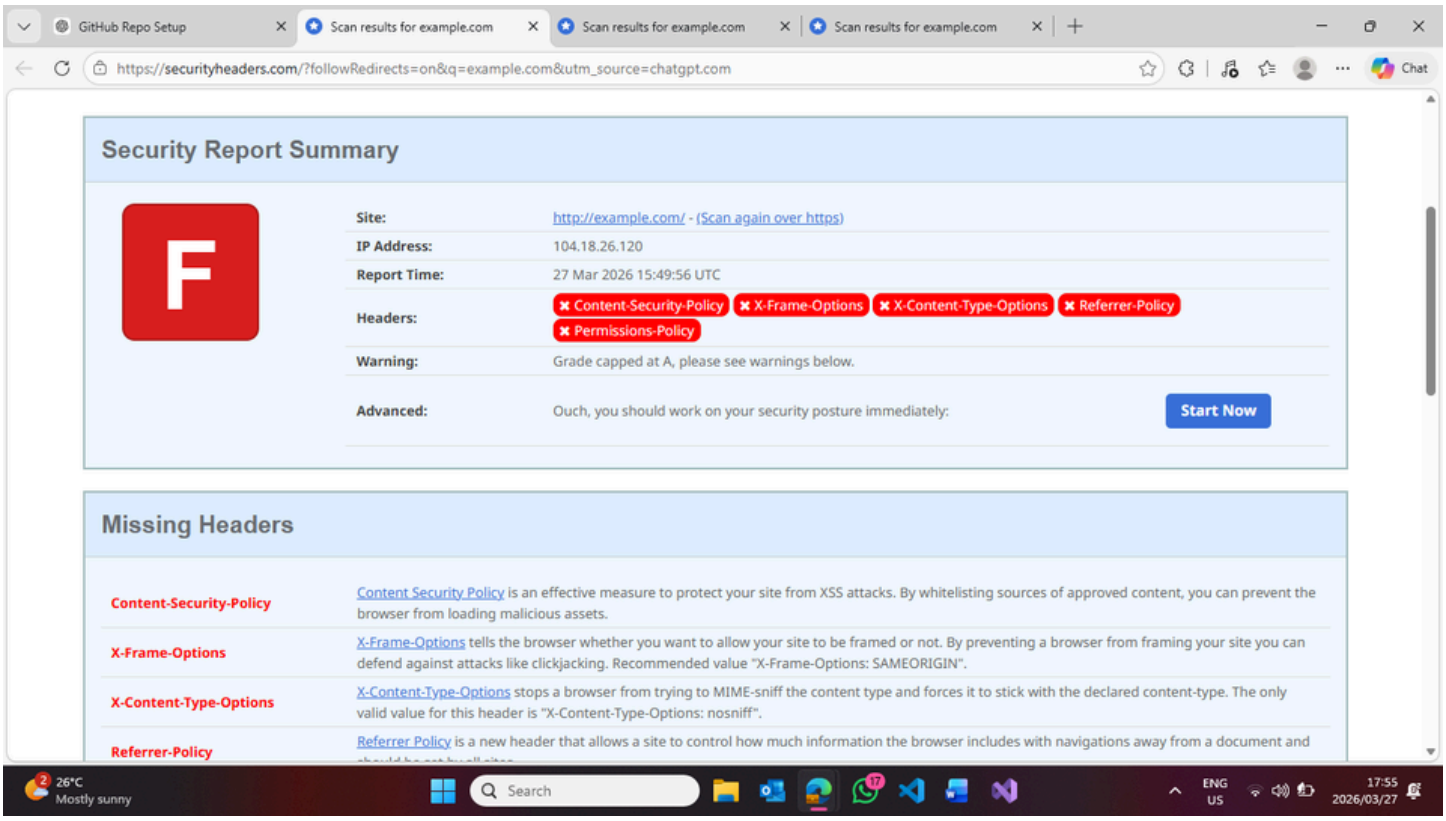
# 4. Findings

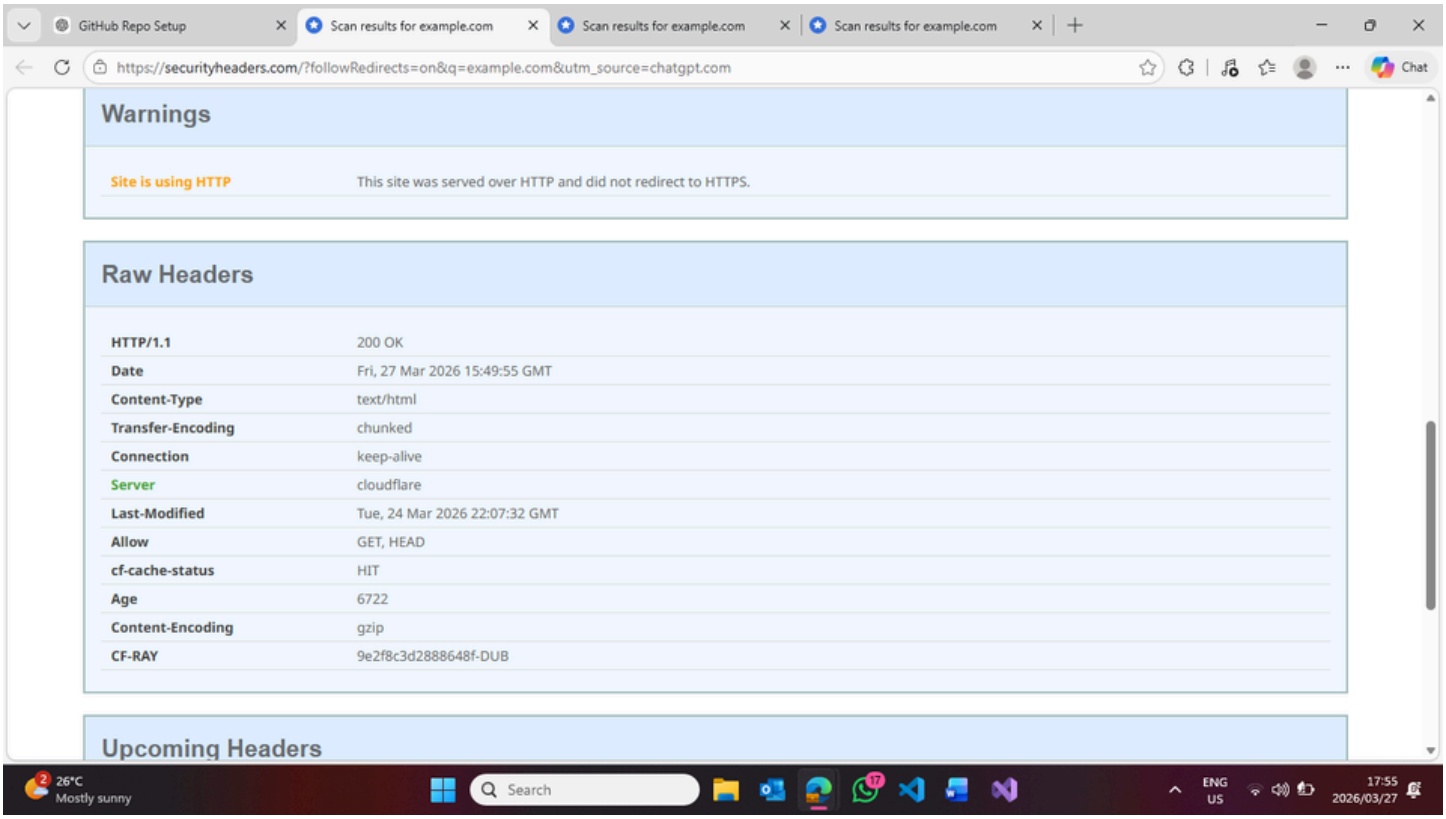
| # | Finding | Description | Why it Matters | Risk Level | Recommendation |
|---|---|---|---|---|---|
| 1 | HTTPS | HTTPS is implemented | Encrypts user data in transit | Low | Continue enforcing HTTPS; monitor certificate validity |
| 2 | Security Headers | Missing Content-Security-Policy, X-Frame-Options, Strict-Transport-Security | Increases risk of clickjacking, XSS, and related attacks | Medium | Add the recommended security headers |
| 3 | Exposed Server Info | Server type/version visible in HTTP headers | Can provide attackers with information to target known vulnerabilities | Low | Hide or restrict server information in headers |
| 4 | Outdated Components | No visible outdated components | Low risk currently, but outdated libraries could introduce vulnerabilities in the future | Low | Monitor and update components regularly |

## 5.Evidence

**Screenshot 1: Security Header Analysis**

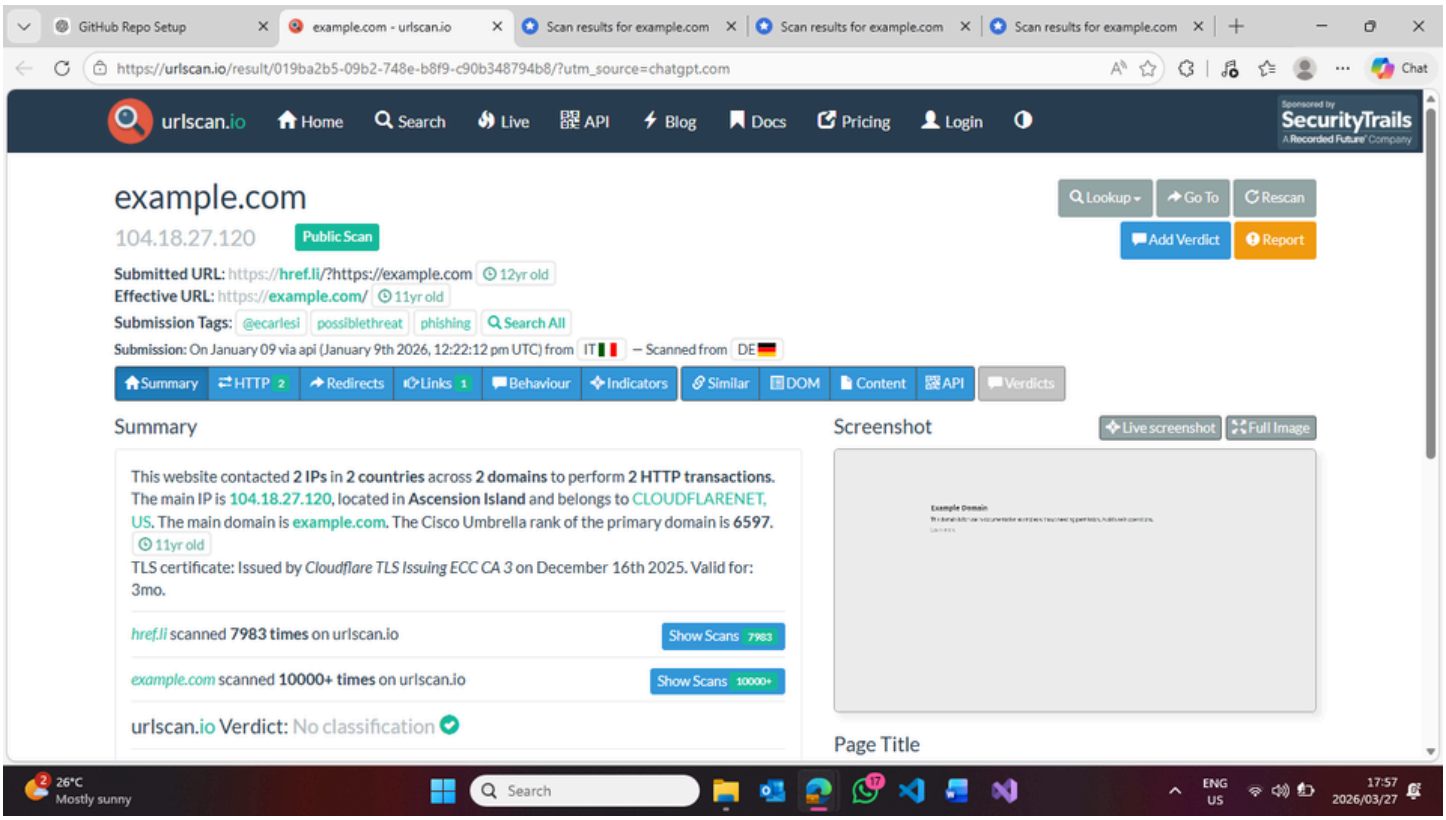
The scan from securityheaders.com shows that several important headers such as Content-Security-Policy, X-Frame-Options, and X-Content-Type-Options are missing for https://example.com. This supports the finding that security headers are not fully implemented.



**Screenshot 2: Browser DevTools – Response Headers**

A passive observation of the response headers shows the presence of Strict-Transport-Security, indicating HTTPS enforcement, but other recommended headers are not present.




# 6. Risk Classification Summary

- **High Risk:** None detected
- **Medium Risk:** Missing security headers
- **Low Risk:** Server info exposure, static components, HTTPS properly implemented

# 7. Recommendations / Remediation

1. **Add Missing Security Headers**
   - Implement Content-Security-Policy, X-Frame-Options, Strict-Transport-Security.
   - Protects against XSS, clickjacking, and related attacks.
2. **Restrict Server Information**
   - Hide server type/version in HTTP headers.
3. **Maintain HTTPS**
   - Ensure SSL/TLS certificates remain valid and HTTPS is enforced on all pages.
4. **Monitor Components**
   - Regularly check and update website components and libraries to prevent future vulnerabilities.

# 8. Conclusion

The website exhibits a low-risk security posture overall, with HTTPS already implemented.

The primary area of improvement is implementing missing security headers, which will significantly reduce exposure to common web attacks.

By addressing these issues, the organization can improve security, protect user data, and maintain trust.